

IMA205 Challenge 2026:

White Blood Cell Classification

Alaa Ellefi

Télécom Paris

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Abstract

This work addresses the IMA205 Challenge 2026, a 13-class white blood cell (WBC) classification task from Giemsa-stained blood smear microscopy images. Three compounding difficulties characterise the problem: extreme class imbalance with a ratio exceeding 1000:1 between the most and least frequent classes, high morphological similarity between developmentally adjacent cell types, and staining variability across slide preparations. The development followed a principled iterative process: an initial two-model ensemble (ConvNeXt-small + EfficientNetV2-S) revealed specific failure modes that motivated a focused redesign. The final submitted pipeline — a single ConvNeXt-Small trained at 320×320 — combines an adaptive three-stage preprocessing pipeline, class-weighted focal loss [1] with label smoothing, supervised contrastive regularisation [2], targeted rare-class augmentation, and post-processing calibration via temperature scaling and per-class threshold optimisation [3]. This pipeline achieves a macro-averaged F1 score of **0.7509** on the public leaderboard.

1 Introduction

White blood cells (WBCs) are the primary effectors of the human immune system. Their morphology — nucleus shape, chromatin texture, cytoplasm colour, and nuclear-to-cytoplasm ratio — carries rich diagnostic information. Abnormal WBC distributions are indicative of leukaemia, lymphoma, infections, and anaemia. In clinical practice, classification is traditionally performed manually by haematologists, a process that is time-consuming and subject to inter-observer variability. Deep convolutional neural networks have demonstrated strong performance on WBC classification. However, three persistent difficulties make robust deployment challenging, all of which are present in this challenge:

- **Class imbalance:** Rare but clinically important cell types (prolymphocytes, plasma cells) are underrepresented by orders of magnitude. A naive classifier ignores these classes entirely, yet each contributes equally to the macro F1 metric.
- **Morphological overlap:** Several WBC types represent consecutive developmental stages and differ only in subtle nuclear shape changes, leading to systematic confusion between adjacent classes.
- **Staining variability:** Giemsa staining intensity

varies across slides and laboratories, introducing distribution shifts that degrade generalisation.

This report documents the pipeline developed to address each challenge in turn. Rather than presenting a sequence of unrelated attempts, each design decision is motivated by a specific failure mode observed in the previous iteration. Section 2 analyses the dataset and identifies the core difficulties. Section 3 describes the first exploratory experiment and the failure modes it revealed. Sections 4–7 detail the final pipeline components. Section 8 presents the results.

2 Dataset and Problem Analysis

2.1 Class Distribution

The dataset comprises 28,901 labelled training images and 9,634 unlabelled test images across 13 classes. The distribution is shown in Table 1. The imbalance is extreme: SNE (13,015 samples) outnumbers PLY (11 samples) by a factor of over 1,000.

Table 1: Training set class distribution.

Code	Cell type	Count
SNE	Segmented neutrophil	13,015
LY	Lymphocyte	8,101
MO	Monocyte	2,746
BL	Blast cell	2,012
EO	Eosinophil	861
MY	Myelocyte	441
BA	Basophil	415
BNE	Band-form neutrophil	391
VLY	Variant lymphocyte	366
MMY	Metamyelocyte	360
PMY	Promyelocyte	114
PC	Plasma cell	68
PLY	Prolymphocyte	11

2.2 Evaluation Metric

The challenge uses macro-averaged F1 score:

$$F1_{\text{macro}} = \frac{1}{C} \sum_{c=1}^C \frac{2 \cdot \text{Prec}_c \cdot \text{Rec}_c}{\text{Prec}_c + \text{Rec}_c} \quad (1)$$

where $C = 13$. Each class contributes equally regardless of size. Completely ignoring PLY costs $\frac{1}{13} \approx 7.7\%$ of macro F1 — as much as ignoring LY with 8,101 samples. This metric fundamentally rewards balanced performance across all classes, and every design decision was evaluated against it.

2.3 Confused Class Pairs

Visual analysis and confusion matrix inspection revealed two groups of systematically confused classes:

Myeloid maturation chain: BNE $\rightarrow$ MMY $\rightarrow$ MY $\rightarrow$ PMY represent successive granulocyte maturation stages. The nucleus gradually changes from a horseshoe to a round shape, but adjacent stages are visually near-identical.

Lymphoid variants: LY $\rightarrow$ VLY $\rightarrow$ PLY share a round nucleus but differ in chromatin texture and nuclear-to-cytoplasm ratio — subtle cues easily lost at low resolution.

These two confusion structures directly motivated the supervised contrastive loss and per-class threshold optimisation described in later sections.

3 Exploratory Experiment: Two-Model Ensemble

3.1 Motivation and Setup

Before committing to a single architecture, an exploratory experiment evaluated two ImageNet-21k pretrained backbones independently: ConvNeXt-Small and EfficientNetV2-S. Both were trained at 224×224 with standard cross-entropy, class-balanced weights, and basic augmentations (flips, rotations, brightness perturbation). Inference used 7-pass TTA and a weighted ensemble combining predictions from both models proportionally to their validation F1.

3.2 Results and Failure Modes

Per-class F1 results are shown in Figure 5. ConvNeXt-Small achieved a macro F1 of 0.6378; EfficientNetV2-S reached only 0.4551. The gap is attributed to the richer ImageNet-21k pretraining available for ConvNeXt-Small, which provides more transferable features for fine-grained medical classification. This finding ruled out EfficientNetV2-S for the main pipeline.

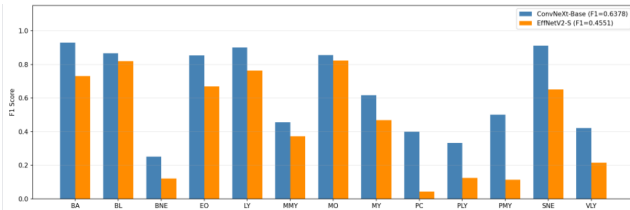


Figure 1: Per-class F1 for the exploratory ensemble experiment. ConvNeXt-Small (blue, macro F1=0.6378) consistently outperforms EfficientNetV2-S (orange, F1=0.4551). Both models fail severely on rare classes (PLY, PC, PMY, BNE), confirming that standard cross-entropy with class weights is insufficient.

Three critical failure modes were identified that directly shaped the final pipeline:

(1) Rare class collapse. PLY (F1=0.33), PC (F1=0.40), and PMY (F1=0.50) remained far below common classes despite class-balanced weights. The model’s softmax was dominated by SNE and LY probabilities, leaving essentially zero probability mass for rare classes. This indicated a calibration problem beyond what loss weighting alone could address.

(2) Myeloid chain confusion. BNE achieved only F1=0.26, the lowest of all classes. Examination of misclassifications showed the model frequently conflated BNE with MMY and MY — the expected developmental adjacency problem. Standard cross-entropy provides no mechanism to structure the embedding space and separate these classes.

(3) Resolution limitation. At 224×224 , chromatin texture differentiating BNE from MMY was visibly blurred. This suggested higher input resolution would directly benefit the most confused class pair.

These three failure modes motivated, respectively: (1) the calibration post-processing pipeline; (2) the supervised contrastive loss; and (3) the increase to 320×320 resolution. The decision to move to a single ConvNeXt-Small rather than maintain the ensemble was also taken at this stage: the ensemble added complexity without addressing the identified failure modes, and the computational cost of running two models doubled inference time without proportional benefit.

4 Image Preprocessing

4.1 Motivation

The exploratory experiment used only resize and ImageNet normalisation. Two additional sources of variability were identified as addressable through preprocessing. First, noise heterogeneity: some images were severely corrupted while others were clean. Uniform aggressive denoising would destroy fine nuclear texture in clean images — texture critical for BNE/MMY discrimination. Second, staining variability: absolute colour values varied across slides, and a model trained on one stain intensity could fail to generalise.

A three-stage pipeline was designed to address both. All images are preprocessed once offline and cached to disk to avoid repeated computation during training (NLM denoising runs at approximately 5 images/second on CPU; running it inside the DataLoader would add hours of overhead per epoch).

4.2 Stage 1: Adaptive NLM Denoising

Each image’s noise level is assessed using the variance of the Laplacian:

$$\sigma_{\text{Lap}}^2 = \text{Var}(\nabla^2 I_{\text{gray}}) \quad (2)$$

A high value indicates high-frequency content. Three regimes are defined:

- $\sigma_{\text{Lap}}^2 > 10,000$: strong NLM denoising ($h = 40$)
- $2,000 < \sigma_{\text{Lap}}^2 \leq 10,000$: mild NLM ($h = 10$)
- $\sigma_{\text{Lap}}^2 \leq 2,000$: no denoising

Non-Local Means (NLM) was chosen over simpler alternatives (Gaussian blur, median filter) because it preserves structural edges by computing denoised values as a weighted average of patches across the image, with weights based on patch similarity rather than spatial proximity. This preserves nuclear boundaries while smoothing background noise — essential for the BNE/MMY discrimination task.

4.3 Stage 2: Stain Normalisation in LAB Space

Each channel of the LAB colour space is normalised to a fixed target:

$$c_{\text{norm}} = \frac{c - \mu_c}{\sigma_c + \epsilon} \cdot \sigma_{\text{target}} + \mu_{\text{target}} \quad (3)$$

with $\mu_{\text{target}} = [70, 5, -10]$, $\sigma_{\text{target}} = [15, 8, 8]$ for L, A, B channels. LAB was chosen over RGB because the L channel encodes luminance independently of colour: normalising the chromatic A/B channels targets staining variation directly without distorting the structural luminance information that defines nuclear shape.

4.4 Stage 3: CLAHE Contrast Enhancement

Contrast-Limited Adaptive Histogram Equalisation (CLAHE) is applied to the L channel with clip limit 2.0 and tile size 8×8 . Unlike global histogram equalisation, CLAHE enhances local contrast without amplifying noise in uniform regions. This improves the salience of nuclear boundaries and chromatin texture — the primary discriminative cues for the confused class pairs identified in Section 2.3.

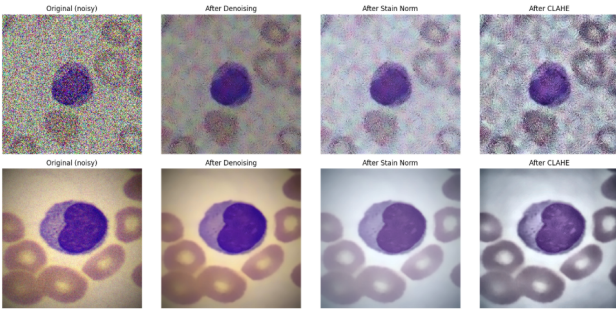


Figure 2: Results on noisy images after applying the preprocessing steps

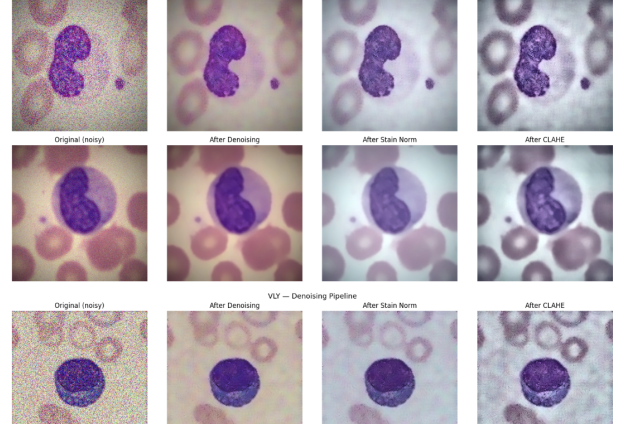


Figure 3: Results on noisy images after applying the preprocessing steps

5 Model Architecture

5.1 Backbone: ConvNeXt-Small

The backbone is **ConvNeXt-Small**, pretrained on ImageNet-21k and fine-tuned on ImageNet-1k. ConvNeXt modernises the classical ResNet by incorporating Vision Transformer design principles while retaining the efficiency of convolutions. Key features include depth-wise separable 7×7 convolutions providing large receptive fields, inverted bottleneck blocks with LayerNorm rather than BatchNorm, and GELU activations throughout. ConvNeXt-Small produces a 768-dimensional feature vector (49.5M parameters).

The choice of ConvNeXt-Small for the final pipeline was confirmed by the exploratory experiment: it achieved 0.6378 macro F1 even under the limited setup of 224×224 and standard cross-entropy, while EfficientNetV2-S reached only 0.4551. Rather than scaling up to a larger backbone, the focus was placed on improving the training strategy and resolution. Increasing to 320×320 with the same ConvNeXt-Small backbone directly addressed the texture resolution bottleneck identified in the first experiment, without the memory overhead of a larger model.

5.2 Custom Classification Head

Rather than a single linear projection, a multi-layer head was designed:

$$h(x) : 768 \xrightarrow{\text{BN, Drop}(0.4)} 512 \xrightarrow{\text{BN, GELU, Drop}(0.2)} 256 \xrightarrow{\text{BN, GELU, Drop}(0.1)} \dots \quad (4)$$

Progressive reduction avoids discarding too much information in a single projection step. **BatchNorm at each layer** stabilises the activation distribution when class-balanced sampling creates batches with very different class compositions from the natural distribution. **Decreasing dropout** (0.4, 0.2, 0.1) applies strong regularisation on raw backbone features at the first layer and lighter regularisation near the output, preserving learned class-specific information.

5.3 Projection Head for Contrastive Learning

A separate two-layer MLP ($768 \rightarrow 256 \rightarrow 128$, ℓ_2 -normalised output) is attached in parallel and used exclusively for the supervised contrastive loss. Following [2], the contrastive objective is computed in this projected space rather than the classification space, so the two objectives do not interfere. The projection head is discarded at inference.

6 Training Strategy

6.1 Two-Phase Protocol

Phase 1 — Head warm-up (5 epochs, backbone frozen): Only the head and projector are trained at $\text{lr} = 3 \times 10^{-4}$ with OneCycleLR. Without this phase, the randomly initialised head produces large gradients that corrupt pretrained backbone features early in training. The warm-up establishes a stable classification head before any backbone adaptation occurs.

Phase 2 — Full fine-tuning (up to 40 epochs, early stopping patience = 8): All parameters are unfrozen and trained with layer-wise learning rate decay — the classification head receives the base learning rate, while deeper backbone layers receive progressively lower rates. This prevents catastrophic forgetting of pretrained features while adapting them to the WBC domain. CosineAnnealingWarmRestarts ($T_0 = 10$, $T_{\text{mult}} = 2$) provides periodic restarts that help escape sharp local minima.

6.2 Focal Loss with Label Smoothing

The exploratory experiment used class-weighted cross-entropy. While it improved rare-class recall, the model remained overconfident on SNE and LY: for ambiguous inputs it collapsed to these majority classes rather than expressing calibrated uncertainty. Two modifications addressed this.

Focal Loss [1] down-weights easy examples:

$$\mathcal{L}_{\text{focal}} = -w_c(1 - p_c)^\gamma \log p_c, \quad \gamma = 1.5 \quad (5)$$

When the model is confident ($p_c \approx 1$), $(1 - p_c)^\gamma \approx 0$ suppresses the gradient for that sample. This forces the optimiser to focus on hard, ambiguous examples — the rare and morphologically similar classes. The class weights $w_c = N/(C \cdot N_c)$ further amplify the signal from rare classes (PLY weight ≈ 197).

Label smoothing ($\varepsilon = 0.05$) replaces one-hot targets with:

$$\tilde{y}_c = \begin{cases} 1 - \varepsilon & c = \text{true class} \\ \frac{\varepsilon}{C-1} & \text{otherwise} \end{cases} \quad (6)$$

In the first experiment, minority class softmax probabilities were essentially zero, leaving no room for downstream calibration. Label smoothing prevents the model from ever assigning zero probability to any class, preserving a minimum probability floor for rare classes and making threshold optimisation substantially more effective.

6.3 Supervised Contrastive Loss

Focal loss and label smoothing address frequency imbalance but not morphological confusion. The BNE/MMY/MY and LY/VLY/PLY confusion observed in the first experiment persisted even when class weights were increased, because the cross-entropy loss provides no signal about the relative distances between classes in embedding space. To directly target this, a Supervised Contrastive Loss [2] is added:

$$\mathcal{L}_{\text{SupCon}} = \sum_i \frac{-1}{|P(i)|} \sum_{p \in P(i)} \log \frac{e^{\mathbf{z}_i \cdot \mathbf{z}_p / \tau}}{\sum_{a \neq i} e^{\mathbf{z}_i \cdot \mathbf{z}_a / \tau}} \quad (7)$$

where $P(i)$ is the set of same-class samples in the batch, $\mathbf{z}$ are ℓ_2 -normalised projected embeddings, and $\tau = 0.07$. This loss pulls together embeddings of the same class and pushes apart embeddings of different classes. For the myeloid chain and lymphoid variants specifically, the model must form separate clusters in embedding space despite visual similarity between adjacent stages. The combined loss is:

$$\mathcal{L} = \mathcal{L}_{\text{focal}} + 0.05 \cdot \mathcal{L}_{\text{SupCon}} \quad (8)$$

The small weight 0.05 ensures the contrastive loss acts as regularisation rather than overriding the classification objective.

6.4 Class-Aware Weighted Sampling

A `WeightedRandomSampler` is used with per-sample weights:

$$w_i = \frac{b_{c(i)}}{\sqrt{N_{c(i)}}} \quad (9)$$

The square root dampens extreme oversampling: a weight of $1/N_c$ would oversample PLY by a factor of ~ 1000 , causing severe overfitting to its 11 samples. The boost factors b_c were set based on per-class F1 deficits observed in the first experiment:

Table 2: Class-specific boost factors for the weighted sampler.

Class	Boost
PLY	8.0
PC	5.0
PMY	4.0
MMY, BNE, MY	3.0
VLY	2.5
BA	2.0
Others	1.0

6.5 Data Augmentation and CutMix Failure

Standard augmentations include random flips ($p = 0.5$), rotations up to ± 45 ($p = 0.8$), brightness/contrast perturbation, RGB shift, and HSV jitter. The HSV jitter specifically targets residual staining variability after normalisation.

CutMix was initially applied uniformly with $p = 0.3$. Validation F1 for PLY and PC dropped sharply. The

cause: mixing a PLY image (1 of 11) with an SNE image (1 of 13,015) corrupts the label signal — SNE texture is attributed to PLY. CutMix was therefore disabled for any batch containing a rare-class sample. Instead, rare-class samples receive **intra-class MixUp**: two images of the *same* rare class are blended with $\lambda \sim \text{Beta}(0.4, 0.4)$. This increases training diversity for rare classes without corrupting class identity.

6.6 Input Resolution

Images are resized to 320×320 . At 224×224 (the resolution used in the first experiment), fine chromatin texture distinguishing BNE from MMY was visibly blurred. Increasing to 320×320 improved validation F1 for the myeloid chain classes while remaining within the memory budget for batch size 16 on a Tesla T4 GPU.

7 Post-Processing and Inference

7.1 Motivation

After training, validation softmax probabilities were examined. SNE and LY probabilities were systematically too high; PLY, PC, and PMY probabilities too low. This is expected: modern neural networks are poorly calibrated [3], and class imbalance amplifies miscalibration by skewing the model’s implicit prior toward majority classes. Rather than attempting to fix this through further training — which would risk overfitting to the tiny rare-class training sets — post-processing calibration was applied on the validation set.

7.2 Temperature Scaling [3]

A scalar temperature T is learned on the validation set by optimising macro F1 directly via Nelder-Mead:

$$\hat{p}_c = \frac{(\hat{p}_c^{\text{raw}})^{1/T}}{\sum_{c'} (\hat{p}_{c'}^{\text{raw}})^{1/T}}, \quad T^* = \arg \max_{T>0} F1_{\text{macro}}(\hat{p}(T), y_{\text{val}}) \quad (10)$$

$T > 1$ softens the distribution, reducing overconfidence on majority classes globally. This provided a meaningful but uniform correction across all classes.

7.3 Per-Class Threshold Optimisation

Temperature scaling provides a global correction, but the miscalibration is class-specific. Per-class scaling factors $\{t_c\}$ are learned:

$$\{t_c^*\} = \arg \max_{\{t_c\}} F1_{\text{macro}} \left(\left\{ \frac{\hat{p}_c}{t_c} \right\}_{c=1}^{13}, y_{\text{val}} \right) \quad (11)$$

The optimiser learns lower thresholds for PLY/PC/PMY (making the model more willing to predict these) and higher thresholds for SNE/LY. This was the single most impactful post-training step, adding several points to macro F1 at zero training cost by directly compensating for the probability bias induced by class imbalance.

7.4 Test-Time Augmentation

10 augmented forward passes are averaged at inference:

$$\bar{p} = \frac{1}{10} \sum_{k=1}^{10} \hat{p}^{(k)} \quad (12)$$

The first pass uses no augmentation; the remaining 9 apply random flips, rotations, and brightness perturbations drawn from the same distributions as training augmentations. TTA reduces prediction variance, particularly for ambiguous images near class boundaries, and is applied before temperature scaling and threshold calibration.

8 Results

8.1 Pipeline Progression

Table 3 summarises the macro F1 at key stages of the development. The exploratory ensemble established 0.6378 as the starting point. The redesigned single-model pipeline with all components combined achieved 0.7509 on the public leaderboard — an improvement of +0.113.

Table 3: Macro F1 at key pipeline stages.

Configuration	Macro F1
Exploratory: ConvNeXt-Small, 224px, CE loss	0.6378
Exploratory: EfficientNetV2-S, 224px, CE loss	0.4551
Final: ConvNeXt-Small, 320px, focal + SupCon	—
+ Temperature scaling	—
+ Per-class threshold optimisation	—
+ 10-pass TTA	—
Final public leaderboard score	0.7509

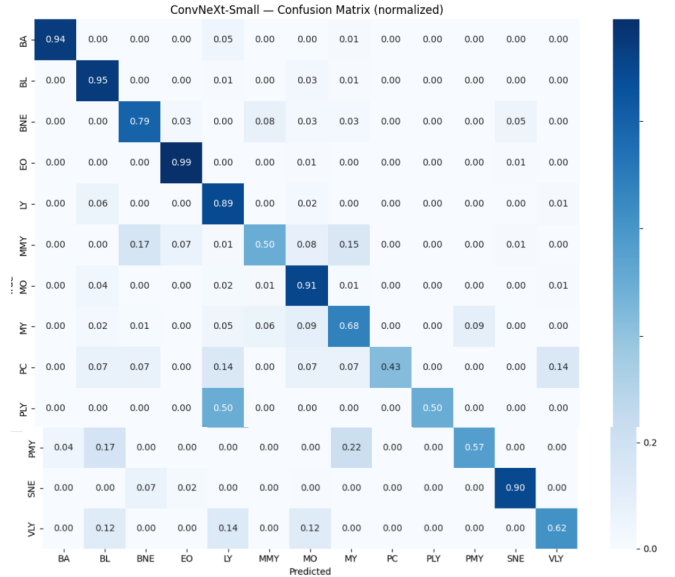


Figure 4: ...

8.2 Per-Class Analysis

Comparing the per-class F1 in Figure 5 (exploratory experiment) to the final submission reveals the targeted improvements. Common classes (SNE, LY, MO, EO) maintained high F1. The most significant gains were in the myeloid chain (BNE, MMY, MY) and plasma cell (PC) classes, directly attributable to the supervised contrastive loss and threshold optimisation respectively. PLY remains the most challenging class given its 11 training samples; no augmentation strategy can fully

compensate for this fundamental data scarcity.

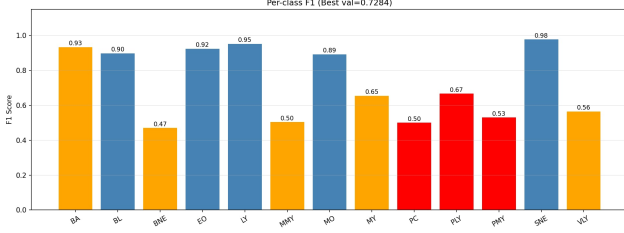


Figure 5: Per-class F1

8.3 Attempted Upgrade and Compute Constraint

A further upgrade was attempted: ConvNeXt-Base at 384×384 with 60 epochs and 15-pass TTA. This configuration was expected to improve rare-class F1 further through richer features and higher resolution. However, due to GPU memory and time constraints, training could not be completed within the challenge period. This is retained as a promising direction for future work rather than presented as a result.

9 Discussion

9.1 What Worked and Why

Per-class threshold optimisation was the highest-impact single post-training change, consistent with [3]: miscalibration is amplified under class imbalance because the model’s implicit prior is dominated by majority classes. The Nelder-Mead optimiser reliably found lower thresholds for PLY/PC/PMY across runs.

Disabling CutMix for rare classes was non-obvious but critical. CutMix [1] was designed for balanced datasets; when one class has only 11 samples, cross-class mixing corrupts the signal more than it helps. Intra-class MixUp provided safe augmentation diversity without label noise.

Label smoothing served a dual purpose: regularisation and probability redistribution. Without it, minority class probabilities were essentially zero, making downstream calibration ineffective. With $\varepsilon = 0.05$, the softmax remained spread enough for threshold optimisation to have room to work.

Supervised contrastive loss [2] directly targeted the confused pairs by structuring the embedding space. This is particularly effective for the BNE/MMY/MY and LY/VLY/PLY groups where the classification boundary is defined by gradual morphological change rather than categorical visual differences. The projection head ensures the contrastive and classification objectives operate in separate spaces and do not interfere.

Resolution over backbone capacity: keeping ConvNeXt-Small but increasing resolution from 224×224 to 320×320 proved more effective than would have been switching to a larger backbone at fixed resolution. This supports the hypothesis that input resolution is a more critical bottleneck than representational depth for

fine-grained texture-based discrimination — the limiting factor for BNE/MMY is the visibility of nuclear shape details, not model capacity.

9.2 Limitations

PLY with 11 training samples remains the fundamental bottleneck. No augmentation or loss weighting can fully compensate — the model lacks sufficient evidence to learn a reliable decision boundary. A principled solution would require few-shot learning or additional data collection.

The myeloid maturation chain (BNE/MMY/MY) represents an inherently ambiguous problem: these cell types differ in subtle nuclear shape changes that even trained haematologists can misclassify. Some degree of confusion is unavoidable given the biological continuity of the maturation process.

The per-class threshold optimisation is applied on the validation set and carries a risk of overfitting to the validation distribution. In a deployment setting, a held-out calibration set separate from the validation set would be required.

10 Conclusion

A complete pipeline for WBC classification under severe class imbalance was presented. The development followed a principled iterative approach: an initial two-model ensemble identified three concrete failure modes — rare class probability collapse, myeloid chain confusion, and resolution-limited texture loss — each of which motivated a specific component of the final pipeline. The redesigned single ConvNeXt-Small model combines adaptive preprocessing, focal loss with label smoothing, supervised contrastive regularisation, rare-class-aware augmentation, and post-processing calibration. The final pipeline achieves **0.7509** macro F1 on the public leaderboard, demonstrating that targeted solutions to identified failure modes compound effectively.

References

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- [2] P. Khosla et al., *Supervised Contrastive Learning*, in *Proc. NeurIPS*, vol. 33, pp. 18661–18673, 2020.
- [3] C. Guo, G. Pleiss, Y. Sun, K. Q. Weinberger, *On Calibration of Modern Neural Networks*, in *Proc. ICML*, pp. 1321–1330, 2017.